Investigate Phase – Axis 5: Brain Load & Hidden Effort

Main Findings and Work Conducted

The Investigate phase aimed to understand how hidden cognitive effort can be detected through functional connectivity using resting-state fMRI.

Research shows that resting brain activity is structured and meaningful.
Marcus E. Raichle demonstrated that spontaneous BOLD signals form organized networks, while Ed Bullmore and Olaf Sporns introduced graph theory to model the brain as interconnected regions.

This supports the idea that dysfunction can exist at the network level even when MRI appears normal.

Key findings:

- Functional imbalance can occur without structural damage
- Hyperconnectivity may reflect compensatory effort
- Network changes can precede visible degeneration

We identified major networks involved in cognitive effort, including the Default Mode, frontoparietal, and salience networks.

We explored graph-based biomarkers such as global efficiency, modularity, and hub centrality to quantify connectivity.

Technically, we defined a pipeline including preprocessing (motion correction, filtering), connectivity matrix extraction, and modeling using statistical and machine learning approaches. We also integrated explainability methods to ensure clinical relevance.

Challenges identified:
data variability, motion artifacts, limited labeled fatigue data, and lack of interpretable models in existing research.

Outcome:
This phase provided a clear scientific and technical framework, identifying key biomarkers, methods, and gaps, and preparing the development of AI-based solutions in the Act phase.